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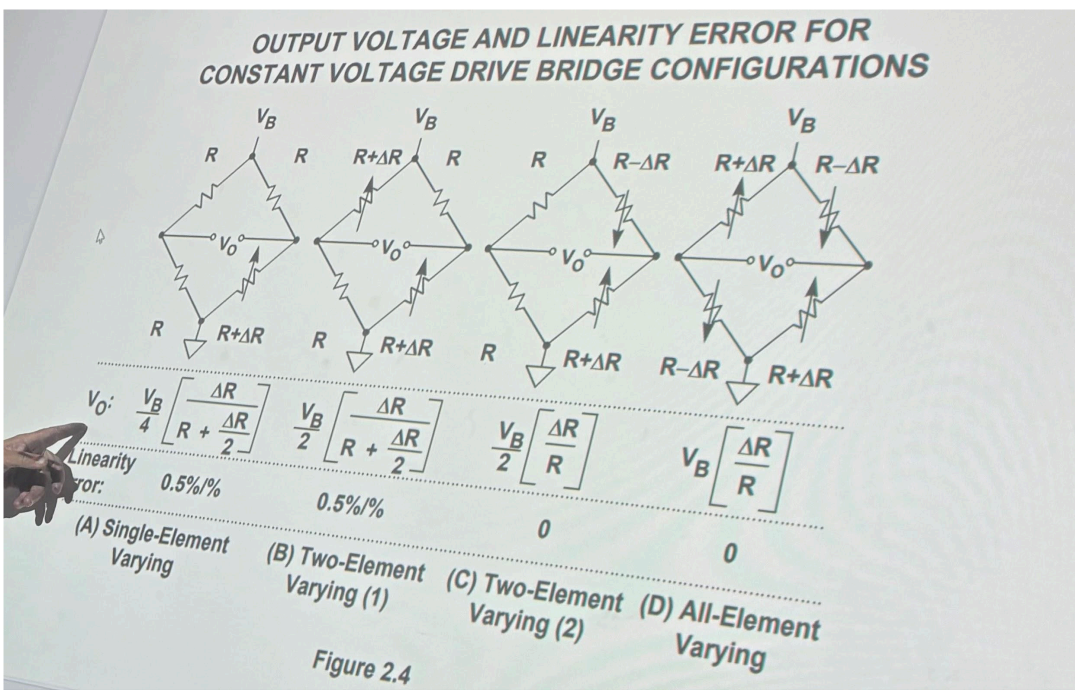
## Introduction

The Wheatstone bridge is a  four resistor network with a DC source. It is so useful because at the balanced (null) condition, the two midpoint voltages are equal, so the meter reads zero. In this state the analysis reduces to two simple voltage dividers and the supply and meter drop out:

$$\frac{Vc}{Vs} = \frac{R3}{R1+r3}\ and\ \frac{VD}{Vs} = \frac{R4}{R2+R4},\ therefore\ Vm = Vc - Vd = 0,\ and\ \frac{R1}{R3} = \frac{R2}{R4},\ and\ Rx = \frac{R3R2}{R1}$$

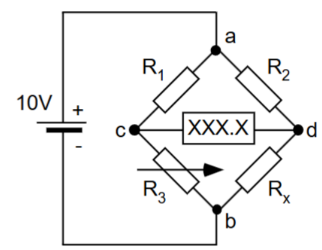
Because Vs cancels and no current flows through the meter at null, the result is insensitive to supply drift and meter resistance—a key advantage both historically and in modern labs. It is important to know that there is no standard or universal labeling for R1, R2, R3 and R4, and mislabling can be a source of unbalance in the arithmetic.There are a few variants of bridges used in this lab- quarter, half and full. Although they use the same ratio at balance, they differ in which resistor changes and therefore in sensitivity near balance.




**IMG 1.** *Wheatstone bridge formulas. These formulas can be used to solve for the voltage between points C and D, whereas in this lab this voltage is mainly measured.*

Quarter-bridge consists of one adjustable resistor and three fixed, known resistor values.Compared to the other configurations, this one has the lowest sensitivity and becomes non-linear once you move far from the null. Because you're not always at balance, supply drift and (to a lesser extent) meter loading can leak into the reading, though modern digital multimeters do help.Half-bridge configurations have two adjustable resistors and two fixed resistors. Full bridge configuration has all four adjustable resistors. It has the lowest linearity error.

## 1. Quarter bridge




**IMG 2.** *Quarter bridge configuration*

First, we chose resistor values in the range of 1k-10k for R1, R2 and Rx. For adjusting our value of R3, we used a resistor box (making sure to not use the comparator top part). The resistors were measured exactly using the EDU34450A digital multimeter. Throughout this lab, EDU34450A digital multimeter and galvanometer was used to measure the voltage between points C and D.

| | R1 | R2 | RX |
|---|---|---|---|
| Theoretical | 5k | 1k | 2k |
| Measured | 5.085k | 0.9990k | 1.997k |

**Table 1.**

The voltage into the circuit was set to 10v, or exactly 10.042v measured going into the circuit.

R3 was adjusted until the voltage measured across points C and D was 0v (or exactly 0.324mv). ***It was found that at a source voltage of 10.04v with a voltage across points C and D of 0.324mv, R3 is approximately 1.58k ohms.*** Then we adjustR3 until the multimeter voltage is 0V, and then we measure and record R3.

The voltage supply was adjusted to show the minimal effect of the source voltage.

Changing the source voltage to 10 volts gives us the following data.

| Theoretical source voltage | Measured source voltage | Measured voltage between point c and d |
| --- | --- | --- |
| +20v | 20.09v | -0.56mv |

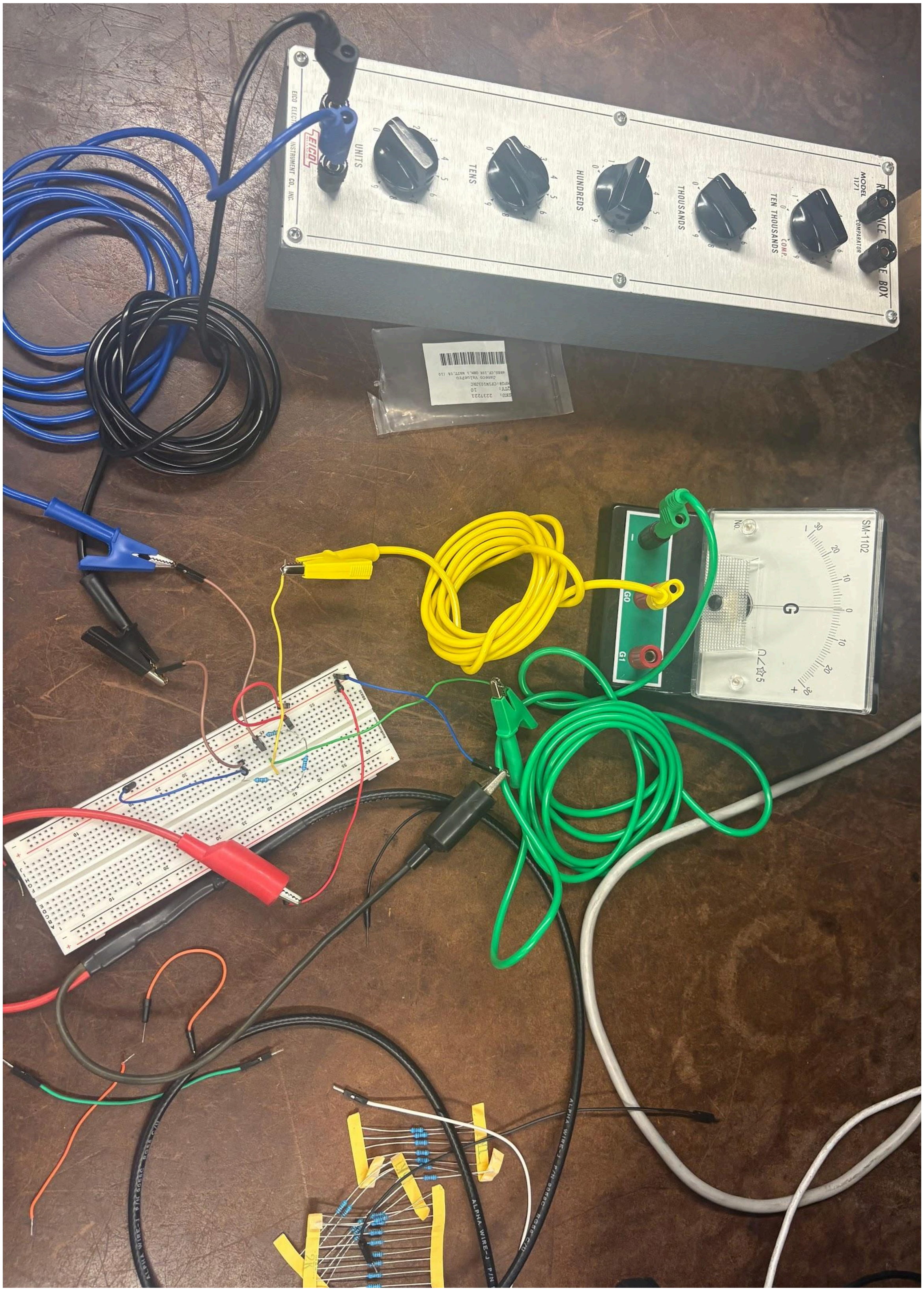
**Table 2.**

Changing the source voltage to one volt gives us the following data.

| Theoretical source voltage | Measured source voltage | Measured voltage between point c and d |
| --- | --- | --- |
| +1v | 1.06v | 0.04mv |

**Table 3.**

As shown, we can see that the voltage stays near 0 for both cases and that the input voltage is not particularly critical or relevant to viewing the effects of the voltage divider formulas of the wheatstone bridges.




**IMG 3.** *Quarter bridge physical circuit.*

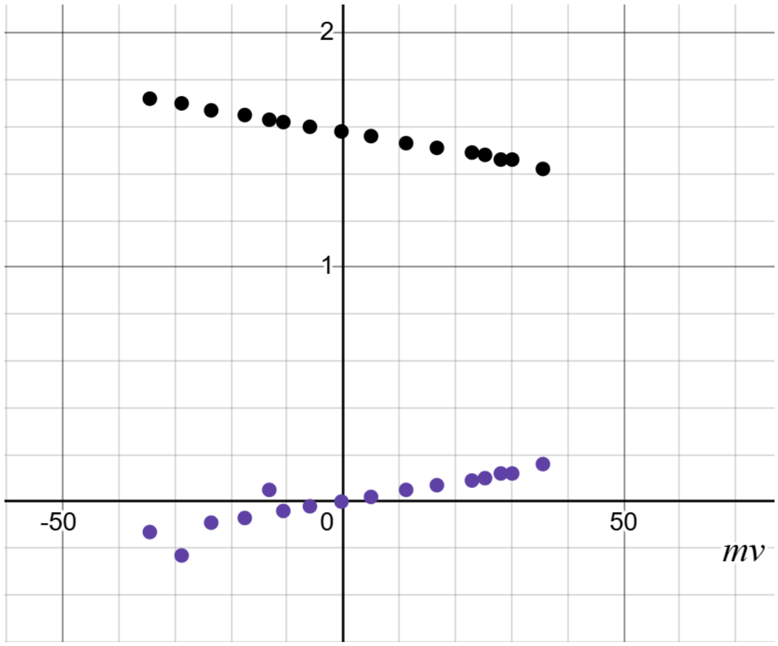
***Quarter bridge operation***

R at balanced condition (0 volts between C and D) was recorded to be 1.58k and $\Delta R = R_{at\ balanced\ condition} - R_{measured}$ So for example, a galvanometer reading of 4mv or digital multimeter reading of 4.96mv,

$\Delta R = R_{at\ balanced\ condition} - R_{measured} = 1.58k - 1.56k = 0.02k$

| R3 (kΩ) | $\Delta R$ (kΩ) | Digital multimeter (mv) | Galvanometer (mv) |
|---|---|---|---|
| 1.42 | 0.16 | 35.6 | 30 |
| 1.46 | 0.12 | 30.12 | 28 |
| 1.46 | 0.12 | 28.1 | 25 |
| 1.48 | 0.1 | 25.28 | 23 |
| 1.49 | 0.09 | 22.95 | 21 |
| `1.51 | 0.07 | 16.7 | 15 |
| 1.53 | 0.05 | 11.21 | 10 |
| 1.56 | 0.02 | 4.96 | 4 |
| 1.58 | 0 | -0.32 | 0 |
| 1.6 | -0.02 | -5.94 | -5 |
| 1.62 | -0.04 | -10.68 | -9 |
| 1.63 | -0.05 | -13.18 | -11 |
| 1.65 | -0.07 | -17.55 | -15 |
| 1.67 | -0.09 | -23.54 | -20 |
| 1.7 | -0.23 | -28.81 | -25 |
| 1.724 | -0.13 | -34.48 | -30 |

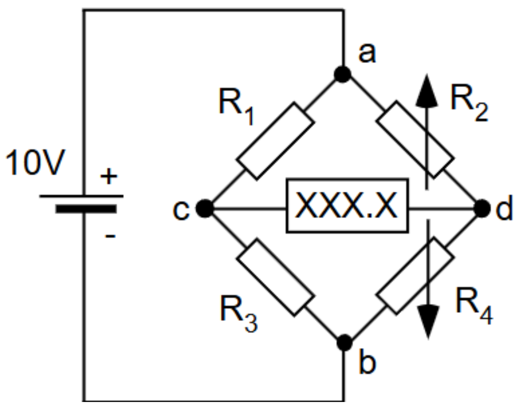
**Table 4.**



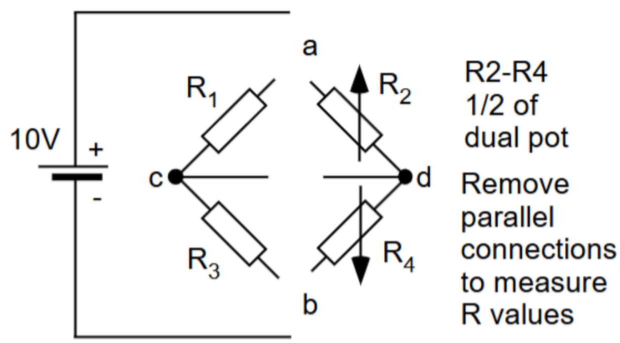

**IMG 4.**

The black points represent the R3 value in k ohms plotted against the voltage between the C and D points. The purple points instead are the change in the resistance. Notice that Rx is 2k, and the resistance is centered around 2k.

## 2. Wheatstone half bridge configuration




**IMG 5.**

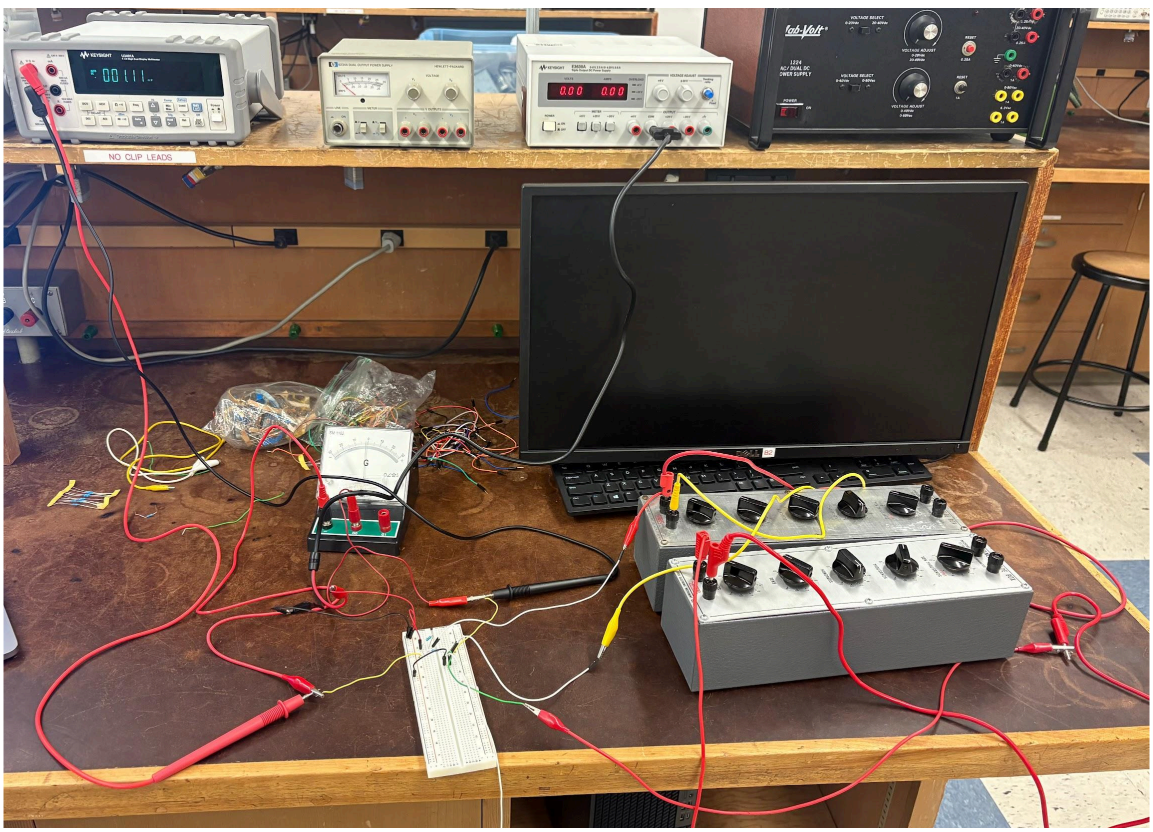



**IMG 6.**

The voltage into the circuit is approximately 10v, as shown below.

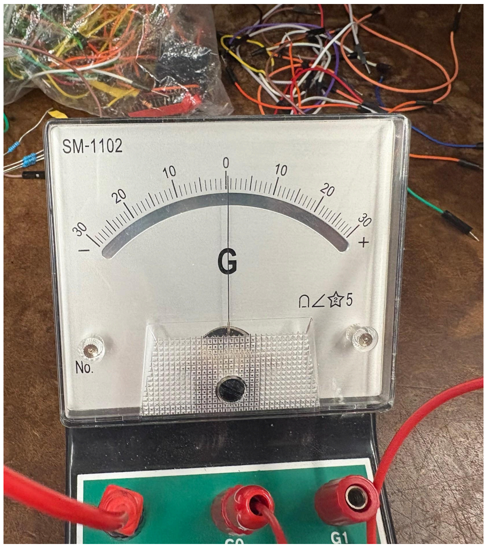

**IMG 7.**

For this configuration, either one variable resistor representing the sum of R2+R4 could be used, or two separate resistor boxes. I chose to use two separate resistor boxes.

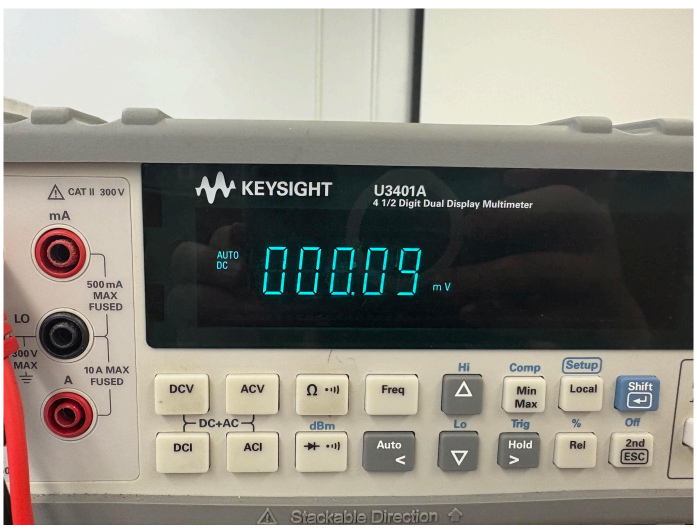



**IMG 8.**

The voltage measured between points C and D, as displayed on the galvanometer and digital multimeter.



**IMG 9.** *Approximately 0v measured by galvanometer*



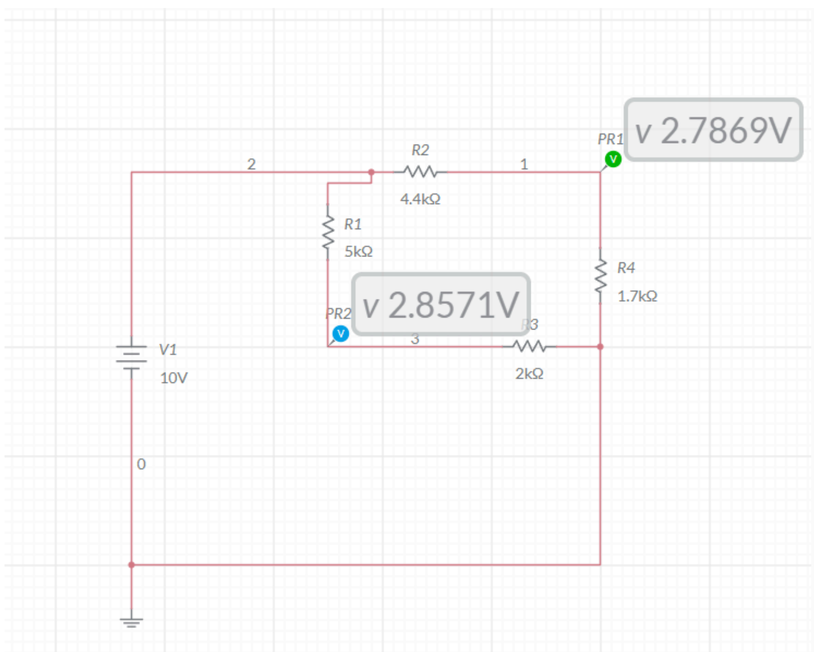
**IMG 10.** *Approximately 0.09mv measured by digital multimeter*

According to image 10, the balanced condition (0v across points C and D) has indeed been met. Again, the bottom part of the resistor boxes were used (not the comparator portion), ***and when measuring R2 and R4 we first disconnected the power supply.*** When trying to achieve the balanced condition, if we raised R2 we would therefore need to decrease R4. In this way we maintain a relationship similar to what we'd have if a potentiometer was used as opposed to this resistor box configuration. It seems the relationship is $\frac{R1}{R3} = \frac{R2}{R4}$, so if $\frac{R1}{R3} = 2.5$ then $\frac{R2}{R4} = 2.5 = \frac{4.4k\Omega}{1.7k\Omega}$. Again, we used resistance amounts between 1-10k ohms.

| | R1 | R3 | R2 | R4 |
|---|---|---|---|---|
| theoretical | 5k | 2k | 4.4k | 1.7k |
| measured | 5.13k | 1.997k | 4.40k | 1.69k |

**Table 5.** *Resistance values at 0.09mv*

Multisim was used to check and get the resistance values ahead of time.

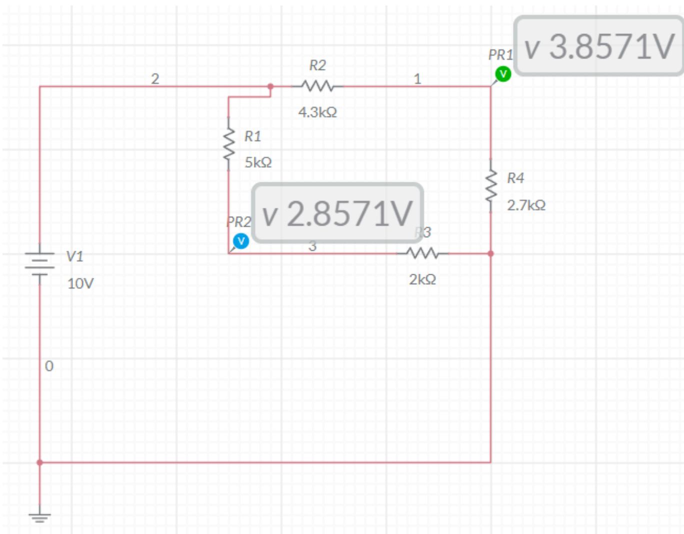


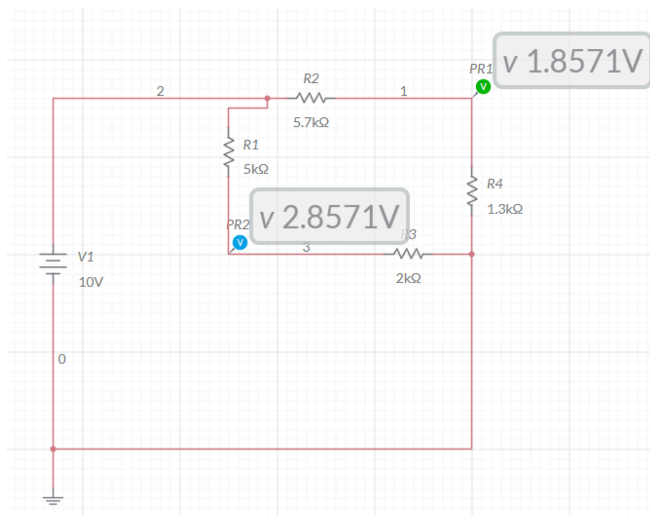

**IMG 11.** *2.8571v-2.7869v = 0.0702v. Therefore the voltage between points C and D is approximately 0v.*

Then, we turn the pot until the meter voltage changes +1V, then re-measure R2 and R4, then repeat for -1v, +2v, and -2v. Mutism was used to find the values of the resistors ahead of time to plug into the physical circuit.

| Voltage across points C and D | R2 measured | R4 measured |
|---|---|---|
| +1v | 4.3k | 2.7k |
| -1v | 5.7k | 1.3k |
| +2v | 3.6k | 3.4k |
| -2v | 9.6k | 0.9k |

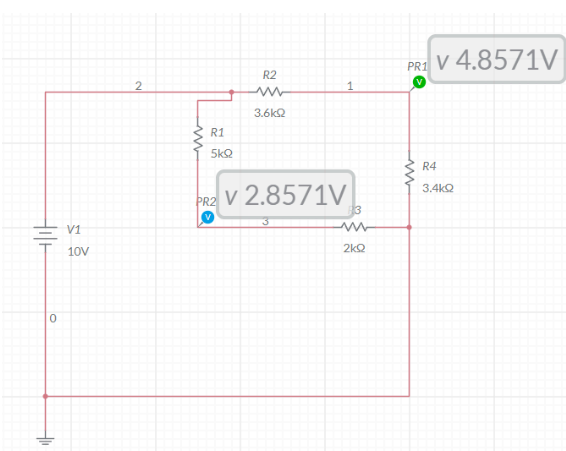
**Table 6.** *Resistance values at various voltages across points C and D*

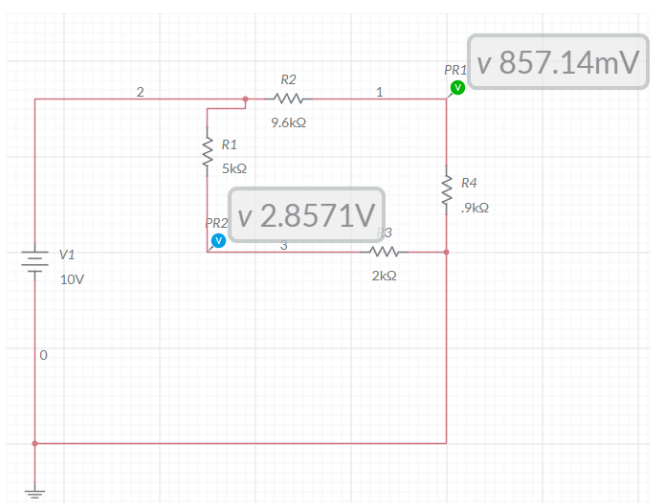



**IMG 12.** *+1V*

**IMG 13.** *-1V*



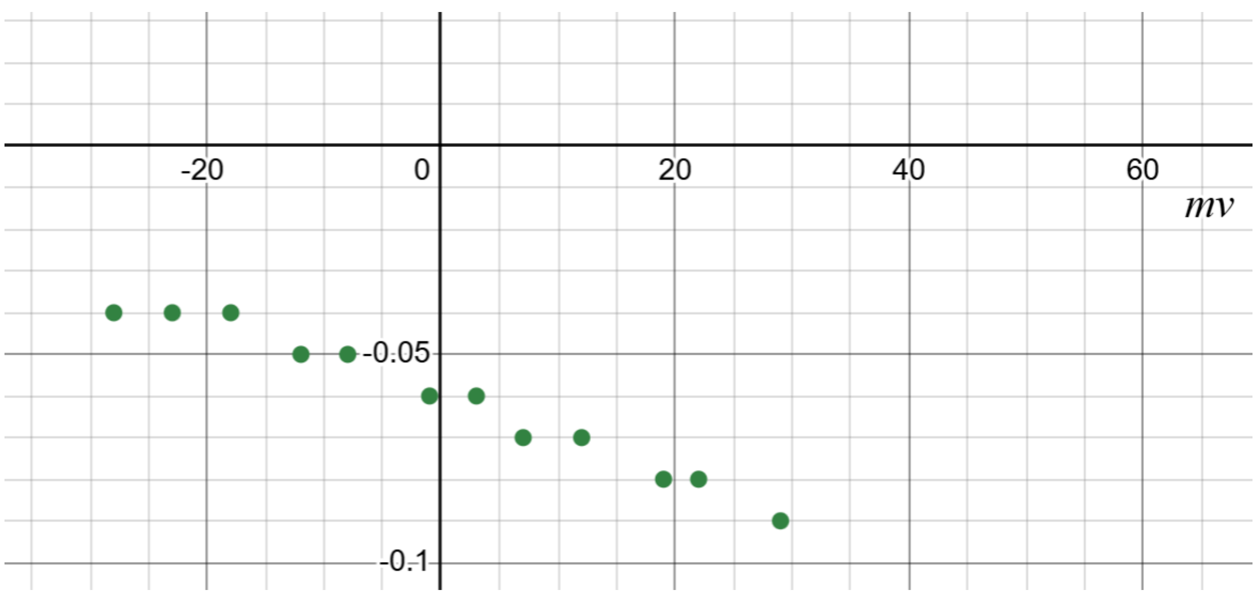

**IMG 14.** *+2v*

**IMG 15.** *-2v*

R4 at balanced condition (0 volts between C and D) was recorded to be 1.7k and as explained previously in the quarter bridge section for table 4, $\Delta R = R_{at\ balanced\ condition} - R_{measured}$ .

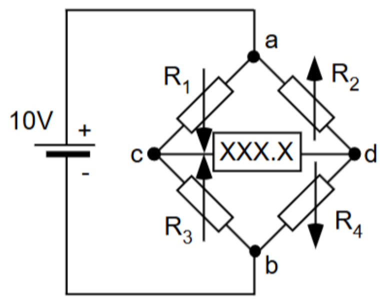
| R4 (kΩ) | $\Delta R$ (kΩ) | Digital multimeter (mv) | Galvanometer (mv) |
|---|---|---|---|
| 1.74 | -0.04 | -33.25 | -28 |
| 1.74 | -0.04 | -26.22 | -23 |
| 1.74 | -0.04 | -20.32 | -18 |
| 1.75 | -0.05 | -13.19 | -12 |
| 1.75 | -0.05 | -10.1 | -8 |
| 1.76 | -0.06 | -2.39 | -1 |
| 1.76 | -0.06 | 4.20 | 3 |
| 1.77 | -0.07 | 9.61 | 7 |
| 1.77 | -0.07 | 13.86 | 12 |
| 1.78 | -0.08 | 21.62 | 19 |
| 1.78 | -0.08 | 25.35 | 22 |
| 1.79 | -0.09 | 33.56 | 29 |

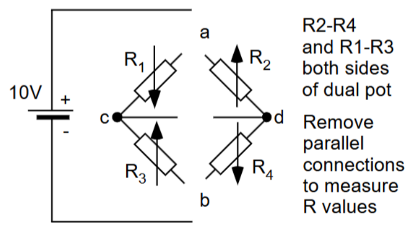
**Table 7.**




**IMG 16.** The points are the change in the resistance plotted against the voltage measurement of Vo.

### 3. Wheatstone full bridge configuration



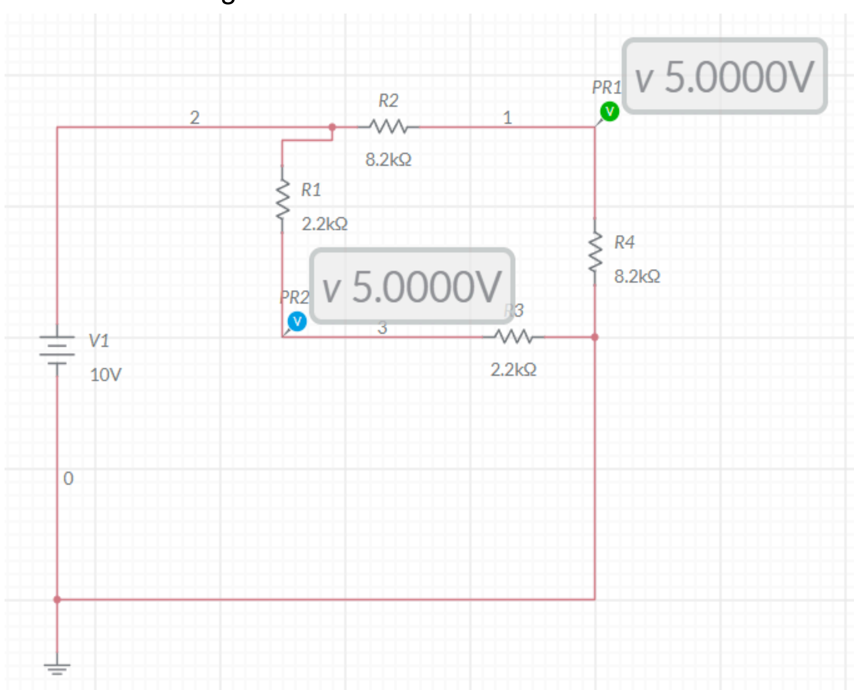

**IMG 17.** **IMG 18.**

Adjust for balanced condition: (0v vout) for 10v

| | R1 | R3 | R2 | R4 |
|---|---|---|---|---|
| theoretical | 2.2k | 2.2k | 8.2k | 8.2k |
| measured | 2.12k | 2.22k | 8.12k | 8.2k |

**Table 8.**

Multisim used to get resistance values needed for balanced conditions.

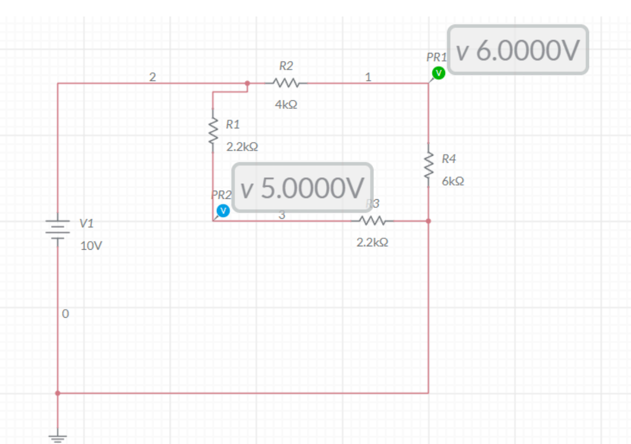



**IMG 19.**

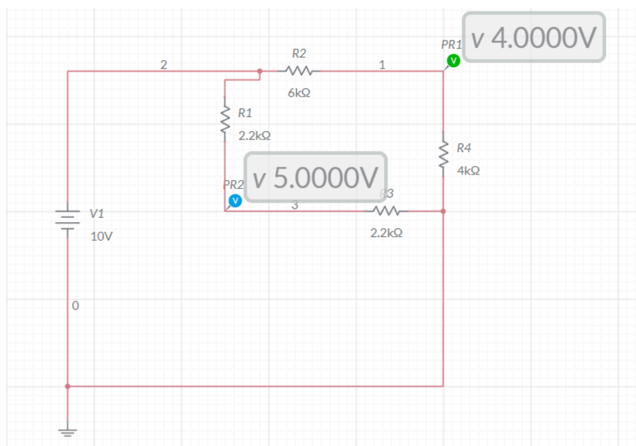
Then, we turn the pot until the meter voltage changes +1V, then re-measure all the adjustable resistances (R1, R2, R3, R4) then repeat for -1v, +2v, and -2v. Mutism was used to find the values of the resistors ahead of time to plug into the physical circuit.

| Voltage across points C and D | R1 | R2 | R3 | R4 |
|---|---|---|---|---|
| +1v | 2.2k | 4k | 2.2k | 6k |
| -1v | 2.2k | 6k | 2.2k | 4k |
| +2v | 2.2k | 3k | 2.2k | 7k |
| -2v | 2.2k | 7k | 2.2k | 3k |

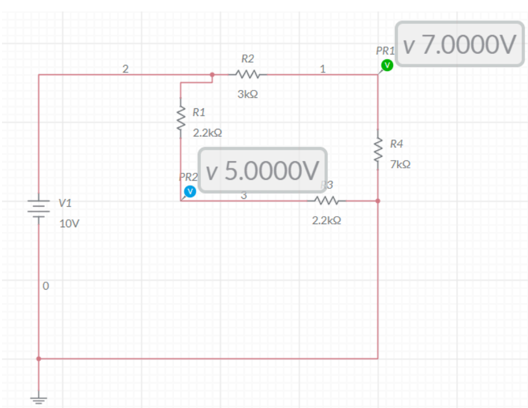
**Table 9.**

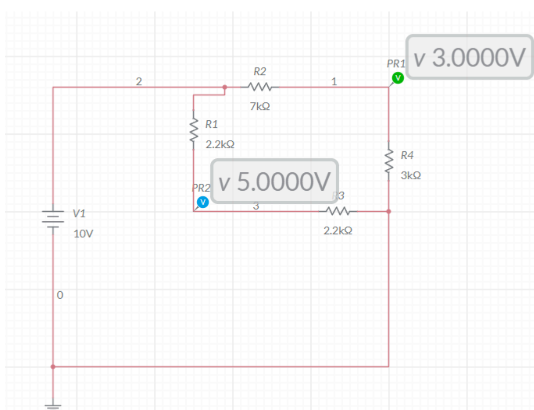



**IMG 20.** *+1V*




**IMG 21.** *-1V*



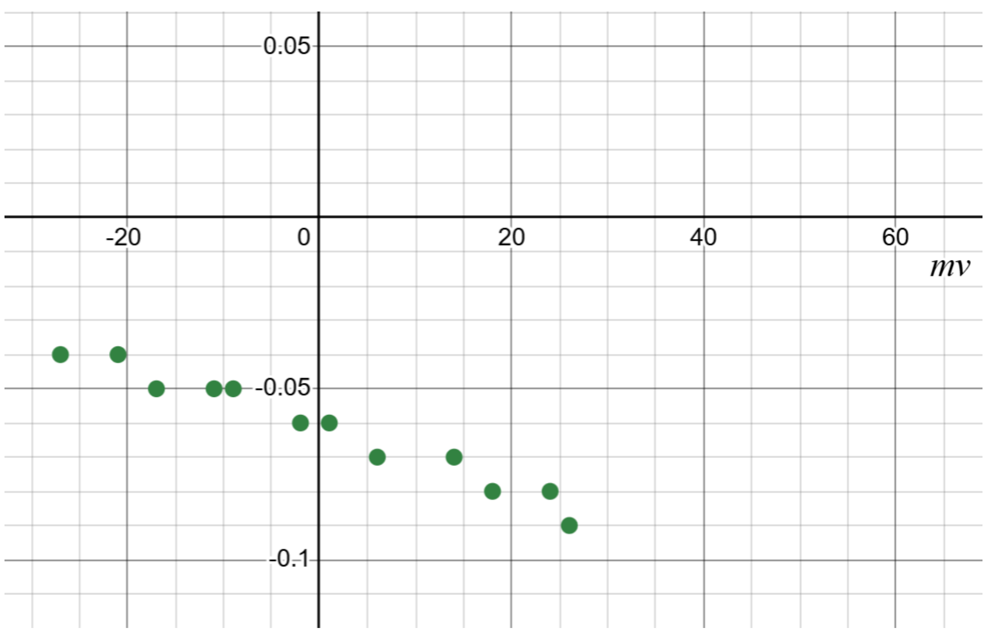

**IMG 22.** *+2V*

**IMG 23.** *-2V*

R4 at balanced condition (0 volts between C and D) was recorded to be 1.7k and as explained previously in the quarter bridge section for table 4, $\Delta R = R_{at\ balanced\ condition} - R_{measured}$.

| R4 (kΩ) | $\Delta R$ (kΩ) | Digital multimeter (mv) | Galvanometer (mv) |
|---|---|---|---|
| 1.74 | -0.04 | -29.10 | -27 |
| 1.74 | -0.04 | -22.30 | -21 |
| 1.75 | -0.05 | -18.8 | -17 |
| 1.75 | -0.05 | -13.00 | -11 |
| 1.75 | -0.05 | -10.5 | -9 |
| 1.76 | -0.06 | -4.22 | -2 |
| 1.76 | -0.06 | 2.16 | 1 |
| 1.77 | -0.07 | 7.98 | 6 |
| 1.77 | -0.07 | 16.34 | 14 |
| 1.78 | -0.08 | 19.23 | 18 |
| 1.78 | -0.08 | 25.76 | 24 |
| 1.78 | -0.08 | 28.19 | 26 |

**Table 10.**




**IMG 24.** The points are the change in the resistance plotted against the voltage measurement of Vo. Very similar to the graph of the half bridge, with a similar amount of linearity and similar linearity error.

**Conclusion**

This lab showed the advantages and usefulness of the Wheatstone bridge circuit and how quarter-, half-, and full-bridge arrangements trade simplicity for sensitivity and linearity error issues. In every case the bridge is balanced when the resistors ratios match (R1/R3=R2/R4), and interestingly enough the measured voltage across points C and D (Vo) remains essentially 0 even when the supply voltage changes from 1v to 20v. This demonstrates the unique concept that the supply voltage is basically unnecessary to include in calculations and in consideration. The quarter-bridge behaved as expected- near the balance point the response was nearly linear, but it became increasingly nonlinear as the resistance moved farther from the balanced condition. The half bridge also behaved as expected- the output became almost linear over a wider range and the sensitivity almost doubled. Finally, the full bridge behaved as expected- similar to the half bridge, with the highest sensitivity/ adjustability, similar linearity to the half bridge but improved from the quarter bridge- requiring smaller resistance changes to reach the same output. Overall, the results align with theory and formulas discussed in the lecture: quarter-bridge is simplest but least sensitive, half bridge (this particular configuration) and full bridge are better then quarter bridge, and wheatstone bridges are a unique case where the voltage source can essentially be disregarded.